

Soru-1 $f(x) = \ln(x-1) + \cos(x-1)$, $f'(x) = \frac{1}{x} - \sin(x-1)$

→ $\ln$ 'in iai negatif çıkamayacağı için, bu fonksiyonlarımız $[1,3,2]$ aralığında sürekli ve türevlenebilir fonksiyon olur.

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ formülünü kullanarak, $x_0 = 1,3$ olsun;

$n=0$ için; $1,3 - \frac{\ln(0,3) + \cos(0,3)}{1/0,3 - \sin(0,3)} \Rightarrow x_1 = 1,38184714$
 $|x_1 - x_0| = 0,08184714 > 10^{-5}$ olduğundan devam

$n=1$ için; $x_2 = x_1 - \left(\frac{f(x_1)}{f'(x_1)} \right) = 1,38184714 - \left(\frac{\ln(0,38184714) + \cos(0,38184714)}{1/0,38184714 - \sin(0,38184714)} \right)$
 $x_2 = 1,397320733$ $|x_2 - x_1| = 0,015473593 > 10^{-5}$ olduğundan devam

$n=2$ için; $x_3 = x_2 - \left(\frac{f(x_2)}{f'(x_2)} \right) = 1,397320733 - \left(\frac{\ln(0,397320733) + \cos(0,397320733)}{1/0,397320733 - \sin(0,397320733)} \right)$
 $x_3 = 1,39774816447$ $|x_3 - x_2| = 0,0004274317 > 10^{-5}$ olduğundan devam

$n=3$ için; $x_4 = x_3 - \left(\frac{f(x_3)}{f'(x_3)} \right) = 1,39774816447 - \left(\frac{\ln(0,39774816447) + \cos(0,39774816447)}{1/0,39774816447 - \sin(0,39774816447)} \right)$
 $x_4 = 1,39774847596$ $|x_4 - x_3| = 0,00000031149 < 10^{-5}$

$x_4 = 1,39774847596$ denklemin istenilen kökü.

Soru-2)

a) $f'(0,4) \cong \frac{1}{2 \cdot (0,2)} [-3f(0,4) + 4 \cdot f(0,6) - f(0,8)]$

$\cong \frac{1}{0,4} [-3(0,9177710) + 4 \cdot (0,8080380) - (0,6386093)]$

$\cong \frac{0,5113137}{0,4} = 1,2782842$

$f'(0,4) \cong [f(0,6) - f(0,2)] \cdot \frac{1}{2 \cdot (0,2)}$

$\cong \frac{1}{0,4} [0,8080380 - 0,9798652] \cong \frac{-0,1718272}{0,4} \cong -0,429568$

$f'(0,4) \cong \frac{1}{12 \cdot (0,2)} [f(0,2) - 8f(0,6) + 8f(0,8) - f(1)]$

$\cong \frac{1}{2,4} \cdot [(0,9798652) - 8 \cdot (0,8080380) + 8 \cdot (0,6386093) - (0,3843735)]$

$\cong \frac{1}{2,4} \cdot (-0,7599373) \cong (-0,3166407)$

$$\begin{aligned}
 f''(0,4) &\cong \frac{1}{(0,2)^2} [f(0,2) - 2f(0,4) + f(0,6)] \\
 &\cong \frac{1}{0,04} [(0,9798652) - 2 \cdot (0,9177710) + (0,8080380)] \\
 &\cong (-1,19097)
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } f'(0,6) &\cong \frac{1}{2 \cdot (0,2)} \cdot [-3f(0,6) + 4 \cdot f(0,8) - f(1)] \\
 &\cong \frac{1}{0,4} \cdot [-3 \cdot (0,8080380) + 4 \cdot (0,6386093) - (0,3843735)] \\
 &\cong \frac{(-0,2540503)}{0,4} = (-0,6351257)
 \end{aligned}$$

$$\begin{aligned}
 f'(0,6) &\cong \frac{1}{2 \cdot (0,2)} \cdot [f(0,8) - f(0,4)] \cong \frac{1}{0,4} \cdot [(0,6386093) - (0,9177710)] \\
 &\cong (-0,6979042)
 \end{aligned}$$

$$\begin{aligned}
 f'(0,6) &\cong \frac{1}{12 \cdot (0,2)} [f(0,2) - 8f(0,4) + 8f(0,8) - f(1)] \\
 &\cong \frac{1}{2,4} [(0,9798652) - 8(0,9177710) + 8 \cdot (0,6386093) - (0,3843735)] \\
 &\cong (-0,6824174)
 \end{aligned}$$

$$\begin{aligned}
 f''(0,6) &\cong \frac{1}{(0,2)^2} [f(0,4) - 2f(0,6) + f(0,8)] \\
 &\cong \frac{1}{0,04} [(0,9177710) - 2 \cdot (0,8080380) + (0,6386093)] \\
 &\cong (-1,4923925)
 \end{aligned}$$

$$\begin{aligned}
 f''(0,6) &\cong \frac{1}{(0,4)^2} [f(0,2) - 2f(0,6) + f(1)] \\
 &\cong \frac{1}{0,16} [(0,9798652) - 2 \cdot (0,8080380) + (0,3843735)] \\
 &\cong (-1,5733831)
 \end{aligned}$$

$$f(x) = 2x + 3\cos x - e^x$$

Soru-3

$$\text{Kirisler Metodu Formül} \Rightarrow x_{n+1} = x_n - \frac{f(x_n) \cdot (x_{n-1} - x_n)}{f(x_{n-1}) - f(x_n)}$$

$$n=0 \Rightarrow x_1 = x_0 - \frac{f(x_0) \cdot (x_1 - x_0)}{f(x_1) - f(x_0)}, \quad x_{-1} = 1 \quad x_0 = 2 \text{ için}$$

$$x_1 = 1,16292513746$$

$$|x_1 - x_0| = 0,83707486254$$

$$n=1 \Rightarrow x_2 = x_1 - \frac{f(x_1) \cdot (x_0 - x_1)}{f(x_0) - f(x_1)}$$

$$x_2 = 1,2164104232$$

$$|x_2 - x_1| = 0,05348528574$$

$$n=2 \Rightarrow x_3 = x_2 - \frac{f(x_2) \cdot (x_1 - x_2)}{f(x_1) - f(x_2)}$$

$$x_3 = 1,24062420207$$

$$|x_3 - x_2| = 0,02421377887$$

$$n=3 \Rightarrow x_4 = x_3 - \frac{f(x_3) \cdot (x_2 - x_3)}{f(x_2) - f(x_3)}$$

$$x_4 = 1,2396841534$$

$$|x_4 - x_3| = 0,00094004867$$

$$n=4 \Rightarrow x_5 = x_4 - \frac{f(x_4) \cdot (x_3 - x_4)}{f(x_3) - f(x_4)}$$

$$x_5 = 1,2397146827$$

$$|x_5 - x_4| = 0,0000305293$$

$$n=5 \Rightarrow x_6 = x_5 - \frac{f(x_5) \cdot (x_4 - x_5)}{f(x_4) - f(x_5)}$$

$$x_6 = 1,23971466742$$

$$|x_6 - x_5| = 0,00000001528 < 10^{-5}$$

denkleminizin $\varepsilon = 10^{-5}$ keskinlikli kökünü $x_6 = 1,23971466742$ bulduk

$f(x) = 2x + 3\cos x - e^x$ işlemini hesap makinesinde bulup,
cevapta direk yerine kaydım.

Soru-4

a) $\frac{b-a}{n} = h$ $\frac{25}{100} = \frac{2-0}{n} \Rightarrow n=8$ tane değer

$$\Delta x = h$$

$$x_i = a + i\Delta x$$

$$x_0 = 0 + 0 \cdot \frac{1}{4} = 0$$

$$x_4 = 0 + 4 \cdot \frac{1}{4} = 1$$

$$x_1 = 0 + 1 \cdot \frac{1}{4} = \frac{1}{4}$$

$$x_5 = 0 + 5 \cdot \frac{1}{4} = \frac{5}{4}$$

$$x_2 = 0 + 2 \cdot \frac{1}{4} = \frac{1}{2}$$

$$x_6 = 0 + 6 \cdot \frac{1}{4} = \frac{3}{2}$$

$$x_3 = 0 + 3 \cdot \frac{1}{4} = \frac{3}{4}$$

$$x_7 = 0 + 7 \cdot \frac{1}{4} = \frac{7}{4}$$

$$x_8 = 0 + 8 \cdot \frac{1}{4} = 2$$

$$\int_a^b f(x) \cdot dx = \frac{h}{2} \left[f(a) + 2 \sum_{j=1}^{n-1} f(x_j) + f(b) \right]$$

$$= \frac{h}{2} \left[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + 2f(x_5) + 2f(x_6) + 2f(x_7) + f(x_8) \right]$$

$$= \frac{1}{8} \left[f(0) + 2f\left(\frac{1}{4}\right) + 2f\left(\frac{1}{2}\right) + 2f\left(\frac{3}{4}\right) + 2f(1) + 2f\left(\frac{5}{4}\right) + 2f\left(\frac{3}{2}\right) + 2f\left(\frac{7}{4}\right) + f(2) \right]$$

$$= \frac{1}{8} \left[0 + 2\left(16 \ln \frac{17}{16}\right) + 2\left(\frac{1}{4} \ln \frac{5}{4}\right) + 2\left(\frac{9}{16} \ln \frac{25}{16}\right) + 2(\ln 2) + 2\left(\frac{25}{16} \ln \frac{41}{16}\right) + 2\left(\frac{9}{4} \ln \frac{18}{4}\right) + 2\left(\frac{49}{16} \ln \frac{65}{16}\right) + 4(\ln 5) \right]$$

$$\cong 3,159475674$$

b) $h = \frac{b-a}{n}$ $\int_a^b f(x) \cdot dx = \frac{h}{3} \left[f(a) + 2 \sum_{j=1}^{\frac{n}{2}-1} f(x_{2j}) + 4 \sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) + f(b) \right]$

$$= \frac{h}{3} \left[f(x_0) + 4(f(x_1) + f(x_3) + f(x_5) + f(x_7)) + 2 \cdot (f(x_2) + f(x_4) + f(x_6)) + f(x_8) \right]$$

$$= \frac{1}{12} \left[0 + 4 \left(\left(\frac{1}{16} \ln \frac{17}{16} \right) + \left(\frac{9}{16} \ln \frac{25}{16} \right) + \left(\frac{25}{16} \ln \frac{41}{16} \right) + \left(\frac{49}{16} \ln \frac{65}{16} \right) \right) + 2 \left(\left(\frac{1}{4} \ln \frac{5}{4} \right) + \ln 2 + \left(\frac{9}{4} \ln \frac{13}{4} \right) + 4 \ln 5 \right) \right]$$

$$\cong 3,109337127$$

$$\begin{aligned}
 \text{c) } h &= \frac{b-a}{n+2} \rightarrow n=6 \quad \int_a^b f(x) \cdot dx = 2h \sum_{j=0}^n f(x_j) \\
 &= 2h [f(x_0) + f(x_2) + f(x_4) + f(x_6)] \\
 &= \frac{1}{2} \left[0 + \left(\frac{1}{4} \ln \frac{5}{4} \right) + \ln 2 + \left(\frac{9}{4} \ln \frac{13}{4} \right) \right] \\
 &\cong 1,700453405
 \end{aligned}$$

Soru-5 Yamuk formülü için hata $|R(h)| \leq \frac{(b-a)^3}{12n^2} M_2$

$$\text{a) } M_2 = \max |f''(x)| \quad a=0 \quad b=2 \quad f(x) = e^{2x} \cdot \sin 3x$$

$$f'(x) = 2e^{2x} \cdot \sin 3x + e^{2x} \cdot 3 \cos 3x$$

$$f''(x) = (4e^{2x} \cdot \sin 3x + 2e^{2x} \cdot 3 \cos 3x) + (2e^{2x} \cdot 3 \cos 3x - e^{2x} \cdot 9 \sin 3x)$$

$$= 12e^{2x} \cdot \cos 3x - 5e^{2x} \cdot \sin 3x$$

$$M_2 = \max |12e^{2x} \cos 3x - 5e^{2x} \cdot \sin 3x| \quad [0, 2]$$

$$x=2 \Rightarrow 648 \cdot (0,960170) - 270 \cdot (-0,279415)$$

$$= 622,13016 + 75,44205 = 697,63221$$

$$|R(h)| \leq \frac{2^3}{12n^2} \cdot 697,63221 < 10^{-4} \quad n^2 \geq \frac{697,63221 \cdot 8 \cdot 10^{-4}}{3^{12}}$$

$$\rightarrow n \geq 0,681967 \quad n=1 \text{ kabul edersek } h = \frac{b-a}{n} = \frac{2}{1} = 2 \quad h=2$$

$$\text{Soru-6 } f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)]$$

$$f'(x) = 3 \cdot e^{3x} - 4 \rightarrow f'(0,5) = 9,4451 \quad x_0 = 0,5 \text{ yerine yazarsak}$$

$$f'(0,5) = \frac{1}{2h} [f(0,5+h) - f(0,5-h)] = 9,4451 \text{ olmalı}$$

$$f(0,5+h) - f(0,5-h) = (18,8902)h$$

$$= e^{3(0,5+h)} - 4(0,5+h) - (e^{3(0,5-h)} - 4(0,5-h)) = (18,8902)h$$

$$= e^{1,5} \cdot e^{3h} - e^{1,5} \cdot e^{-3h} - 8h = (18,8902)h$$

$$= e^{1,5} (e^{3h} - e^{-3h}) = (26,8902)h \Rightarrow e^{3h} - e^{-3h} = 6h$$

$$h \cong 0,0019$$